INSULIN ASPART

High-alert drug

Inject with meals

CHEMICAL NAME B28-aspartic acid human insulin (recombinant insulin analogue)
CLASS Rapid-acting insulin analogue
TRADE NAMES NovoRapid · NovoLog · Fiasp (faster aspart)

THERAPEUTIC USES / INDICATIONS

- Prandial (mealtime) insulin in type 1 and type 2 diabetes (basal-bolus).
- Correction doses for high glucose.
- Insulin pumps (CSII).
- IV use in hospital (where approved).

PHARMACODYNAMICS

- Same actions as human insulin with faster, shorter profile → better post-meal control, fewer late hypos.

ADMINISTRATION

- **SC** 0–15 min before meals (Fiasp: at start of meal or within 20 min after).
- Can be given immediately after meals in children with unpredictable eating.
- Pumps: continuous SC infusion.
- Clear solution.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site reactions, lipohypertrophy.
- Weight gain.

ADVERSE EFFECTS (SERIOUS)

- Hypoglycaemia if injected without eating.
- Hypokalaemia.
- Ketoacidosis rapidly if pump fails (no long-acting depot).

DOSAGES

- **Individualised:** total insulin usually 0.5–1.0 unit/kg/day, of which 50–70% may be NovoRapid at meals.
- Carbohydrate counting: individual insulin-to-carbohydrate ratio set by the diabetes team.
- Correction: per insulin-sensitivity factor.

CONTRAINDICATIONS & CAUTIONS

- Hypoglycaemia; hypersensitivity.

KEY INTERACTIONS

- As other insulins (β-blockers, steroids, alcohol).

PHARMACOKINETICS

- **Onset:** 10–20 min. **Peak:** 1–3 h. **Duration:** 3–5 h.
- Amino-acid substitution prevents hexamer formation → faster absorption.

NURSING MANAGEMENT

- Give when meal tray is present — food within 10–15 min.
- Check glucose before meals.
- Hold or reduce if patient not eating (seek advice).
- Pump patients: check site, ketones if glucose high.

PATIENT EDUCATION

- Inject just before eating; never without food.
- Learn carbohydrate counting and correction doses.
- Carry hypo treatment.
- Pump users: have backup pens.

KEY POINTS

- Rapid-acting analogue — onset 10–20 min.
- Give with meals.
- Basal-bolus and pumps.
- Food must follow the dose.

SOURCES CHECKED **[591]** NovoRapid FlexPen (UK SmPC) **[592]** Insulin Aspart, IV, SC (US label) **[593]** Insulin Aspart, IV, SC (US label, A-S Medication Solutions)

Full references p. 432–457